Undergraduate Research Report

Tripartite Quantum Entanglement and Nonlocality: A Study of One-Extendibility on Alice's Side

Student ID: C24129014

Yu-Hsuan Shih

Faculty Advisor:

Professor Yung-Cheng Liang

National Cheng Kung University

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Abstract

This study asks whether bipartite quantum nonlocality can coexist with extendibility in a tripartite Bell scenario. A third party C acts as a candidate copy of Bob. An ordinary extension requires $p_{AB} = p_{AC}$, and a perfect extension further requires $b = c$ whenever Bob and Charlie use the same setting.

We first reproduce the maximum Bell violations under ordinary and perfect extensions reported in Ref. [6], using our own implementation of linear programs and moment-matrix relaxations. We also show that, for I_{3322}, the NPA level-2 values in that work are reproduced only when the nonnegativity of tripartite probabilities is not imposed; with this constraint, the level-2 bound with ordinary extension drops from 0.125 to 0.0946. A random search over 1,000 three-qubit configurations finds no nonlocal example, but a targeted optimization yields an explicit realization of three qubits with $p_{AB} = p_{AC}$ and $I_{3322} \approx 0.0161 > 0$. Using exact rational arithmetic, we prove that every quantum extension of this p_{AB} has a mean mismatch greater than 0.0963, whereas the same p_{AB} admits an exact perfect no-signaling extension. Thus nonlocality together with an ordinary quantum extension does not imply a perfect quantum extension, and whether perfect copying is possible depends on the theory allowed for the third party. We also study how white noise and a tolerated mismatch affect this separation.

1 Research Objectives

This project concerns the verification of quantum nonlocality under specific conditions. The goal of this research is to determine whether the following question can be proven or generalized:

> *Given that $p(a, b \mid x, y)$ is nonlocal and one-extendible on Alice's side, is it always possible to find a distribution $\{p(a, b, c \mid x, y, z)\}$ such that*
>
> $$p(a, b, c \mid x, y, y) = \delta_{bc}\, p(a, b, c \mid x, y, y)\,? \tag{1}$$

My advisor proposed this problem with the hope that I would read extensively in the research literature and study independently through open courseware, and ultimately see whether I could develop a theoretical derivation that answers it. Therefore, I began by reading introductory textbooks on quantum information, along with the papers my advisor provided. In the process, I realized that the problem is closely related to cryptography. In this setting, the case $b = c$ means that the receiver (party B) and the eavesdropper (party C) always obtain identical results, which would call into question the security of the entire system. Conversely, if this condition cannot be satisfied, the eavesdropper cannot perfectly reproduce the receiver's outcomes, and the security of the system can be established.

I then designed the research plan. The first step was to run numerical simulations to determine whether quantum nonlocality can arise under these conditions. If it can, the next step

would be to attempt a rigorous mathematical analysis; if not, the project would conclude at that point. Due to time constraints and a subsequent change of research topic, the project was left incomplete and did not reach a final conclusion. Nevertheless, this report describes in detail what I have learned and explains the reasoning behind the research plan.

2 Theoretical Background

As research on multipartite quantum nonlocality has advanced, the extendibility of quantum states and the related notion of symmetric extensions have become key tools for understanding the boundaries between entanglement, locality, and nonlocality.

2.1 Extendibility and Its Physical Meaning

In quantum information theory, extendibility describes whether a quantum state can be symmetrically embedded into a larger system. A bipartite state ρ_{AB} is *one-extendible* on Bob's side if there exists a tripartite state $\rho_{ABB'}$, symmetric under the exchange of B and B', that reproduces ρ_{AB} on both pairs:

$$\mathrm{Tr}_{B'}(\rho_{ABB'}) = \rho_{AB}, \qquad \mathrm{Tr}_{B}(\rho_{ABB'}) = \rho_{AB'}, \tag{2}$$

where $\rho_{AB'}$ is identical to ρ_{AB} with B replaced by B'. This is the simplest case of a symmetric extension.

More generally, if ρ_{AB} can be extended to k symmetric copies of B, it is said to have a k-symmetric extension. Doherty, Parrilo, and Spedalieri (2002, 2004) used this hierarchy to construct a complete test for entanglement: a state has a symmetric extension for every k if and only if it is separable. Terhal, Doherty, and Schwab (2003) further showed that if a state has a k-symmetric extension, then any Bell experiment in which Bob uses at most k measurement settings admits a local hidden variable (LHV) model.

Physically, extendibility means that the correlations Alice shares with Bob can be shared equally with additional copies of Bob. This symmetric shareability plays a central role in the security of quantum key distribution, the classification of entanglement, and criteria for quantum nonlocality.

2.2 One-Extendibility on Alice's Side and Tripartite Nonlocality

We now apply this idea to a tripartite system shared by Alice, Bob, and Charlie. A state ρ_{ABC} is one-extendible on Alice's side if there exists a four-party state $\rho_{AA'BC}$, symmetric under the exchange of A and A', such that

$$\mathrm{Tr}_{A'}(\rho_{AA'BC}) = \rho_{ABC}, \qquad \mathrm{Tr}_{A}(\rho_{AA'BC}) = \rho_{A'BC}. \tag{3}$$

In this case, all correlations involving Alice's outcomes can be embedded in a larger symmetric model. This has three main consequences:

1. **Constrained entanglement structure.** One-extendibility on Alice's side limits how strongly Alice can be entangled with the rest of the system. Even when both Bob and Charlie are correlated with Alice, these correlations remain constrained by bipartite locality and separability bounds.

2. **Upper bound on tripartite Bell violation.** Tripartite systems that are one-extendible on Alice's side cannot exceed a strict upper bound in Bell tests involving Alice's measurements. In particular, they cannot reach the strong violations achieved by GHZ-type states, which exhibit maximal tripartite nonlocality.

3. **Monogamy and information security.** This restriction reflects the monogamy of nonlocality: if Alice and Bob exhibit nonlocal correlations, Alice's correlations with Charlie are limited, so the two pairs cannot both be strongly nonlocal.

3 Research Methodology and Design

3.1 Tripartite Quantum System

We consider three spatially separated parties, Alice, Bob, and Charlie. Each party independently chooses a measurement setting (input) and obtains a measurement outcome (output):

$$x, y, z \in \{0, 1, 2\}, \qquad a, b, c \in \{0, 1\}. \tag{4}$$

This is a standard setup for multipartite Bell tests. Using three parties rather than two allows us to study multipartite entanglement and its violation of local bounds. Three settings per party are enough to include common inequalities such as the Mermin and Svetlichny inequalities, and binary outcomes are compatible with the standard GHZ and CHSH frameworks.

The central object is the tripartite conditional distribution

$$p(a, b, c \mid x, y, z). \tag{5}$$

We require the Alice–Bob and Alice–Charlie marginals to be identical,

$$p_{AB} = p_{AC}, \tag{6}$$

and, for perfect extendibility, we additionally require that Bob and Charlie always agree when they use the same setting:

$$p(a, b, c \mid x, y, y) = \delta_{bc}\, p(a, b, c \mid x, y, y). \tag{7}$$

These conditions define the problem precisely and connect it directly to standard tools in multipartite quantum information, such as the NPA hierarchy and the device-independent framework.

3.2 Ordinary and Perfect Extensions

Let m be the number of settings per party. A tripartite distribution $P(a, b, c \mid x, y, z)$ is an *ordinary extension* of p_{AB} if its Alice–Bob and Alice–Charlie marginals both equal p_{AB}. It is a *perfect extension* if, in addition, Bob and Charlie never disagree when they use the same setting. We quantify the disagreement by

$$e_y = P(b \neq c \mid y, z = y), \qquad \bar{e} = \frac{1}{m} \sum_y e_y, \tag{8}$$

so that a perfect extension is equivalent to $\bar{e} = 0$. Nonlocality is tested with the Collins–Gisin family I_{mm22} [**?**], normalized so that its local bound is 0 ($m = 2$ gives CHSH and $m = 3$ gives I_{3322}), and with the CGLMP inequality [**?**] for d outcomes, rescaled as $\frac{d-1}{2d}(I_d - 2)$ so that its local bound is also 0.

3.3 Random Quantum State Generation

To explore tripartite entanglement without bias, we sample pure states $|\psi_{ABC}\rangle$ from the Haar measure. Haar-random states are uniformly distributed over the Hilbert space, so the search is not restricted to states with special symmetries or structures.

To enforce the symmetry between Bob and Charlie, each sampled state is projected onto the subspace symmetric under the exchange of B and C:

$$|\tilde{\psi}_{ABC}\rangle = \frac{(\mathbb{I}_A \otimes \mathbb{I}_{BC} + \mathbb{I}_A \otimes S_{BC}) |\psi_{ABC}\rangle}{\|(\mathbb{I}_A \otimes \mathbb{I}_{BC} + \mathbb{I}_A \otimes S_{BC}) |\psi_{ABC}\rangle\|}, \tag{9}$$

where $S_{BC} = \sum_{i,j} |ij\rangle \langle ji|$ is the swap operator between Bob and Charlie. Combined with identical measurements for Bob and Charlie, this guarantees that tracing out either party yields the same bipartite distribution.

3.4 Quantum Measurement Operators

To avoid bias from any particular choice of settings, each party's measurements are generated randomly. For each of Alice's settings x, we draw a Haar-random unitary U_x and define the projectors

$$P_{a|x} = |a_x\rangle \langle a_x| , \tag{10}$$

where $\{|a_x\rangle\}$ is the orthonormal basis given by the columns of U_x. These projectors automatically satisfy

$$P_{a|x}P_{a'|x} = \delta_{aa'}P_{a|x}, \qquad \text{(orthogonality)} \tag{11}$$

$$\sum_a P_{a|x} = \mathbb{I}. \qquad \text{(completeness)} \tag{12}$$

Bob's and Charlie's projectors $P_{b|y}$ and $P_{c|z}$ are constructed in the same way, with Bob and Charlie using identical operators.

3.5 Quantum Correlation Calculation

Given the state and measurements, the Born rule gives the tripartite distribution

$$p(a, b, c \mid x, y, z) = \langle\tilde{\psi}_{ABC}| P_{a|x} \otimes P_{b|y} \otimes P_{c|z} |\tilde{\psi}_{ABC}\rangle . \tag{13}$$

The corresponding code is shown below.

3.6 Physical Constraints

A valid distribution must satisfy the following conditions.

Normalization and non-negativity.

$$\sum_{a,b,c} p(a, b, c \mid x, y, z) = 1, \qquad p(a, b, c \mid x, y, z) \geq 0. \tag{14}$$

Marginal symmetry. Equation (6), $p_{AB} = p_{AC}$.

No-signaling. For all values of the remaining variables,

$$\sum_a p(a, b, c \mid x, y, z) = \sum_a p(a, b, c \mid x', y, z), \tag{15}$$

$$\sum_b p(a, b, c \mid x, y, z) = \sum_b p(a, b, c \mid x, y', z), \tag{16}$$

$$\sum_c p(a, b, c \mid x, y, z) = \sum_c p(a, b, c \mid x, y, z'). \tag{17}$$

Perfect extension. Equation (7), which is required only when testing perfect extendibility.

3.7 Local Polytope Analysis

The local polytope $\mathcal{L}$ is the set of all distributions that admit a local hidden variable model. Every point in $\mathcal{L}$ is a convex combination of deterministic strategies:

$$\vec{p} = \sum_{\alpha} q_\alpha \vec{d}_\alpha, \qquad q_\alpha \geq 0, \qquad \sum_{\alpha} q_\alpha = 1, \tag{18}$$

where each $\vec{d}_\alpha$ assigns a fixed output to every input. There are finitely many such strategies, and they form the vertices of $\mathcal{L}$. Every Bell inequality corresponds to a facet of this polytope, so a distribution is local if and only if it lies inside $\mathcal{L}$.

To test whether p_{AB} is local, we solve the linear-programming feasibility problem

$$\text{find } q_\alpha \geq 0 \quad \text{such that} \quad \sum_{\alpha} q_\alpha = 1, \qquad \vec{p}_{AB} = \sum_{\alpha} q_\alpha \vec{d}_\alpha. \tag{19}$$

If the problem is feasible, p_{AB} is local; if it is infeasible, p_{AB} is nonlocal.

3.8 Semidefinite Programming and NPA Hierarchy

A distribution that satisfies all of the constraints above, including no-signaling, is not necessarily achievable in quantum mechanics: some "supra-quantum" correlations obey every no-signaling constraint yet have no quantum realization. The Navascués–Pironio–Acín (NPA) hierarchy addresses this problem. It is a sequence of outer approximations to the quantum set that become increasingly tight at higher levels, and each level can be tested as a semidefinite program (SDP).

Given the number of parties, inputs, and outputs, each level defines a finite set of operator products and a corresponding moment matrix. The test asks whether there exists a positive semidefinite moment matrix whose entries are consistent with the observed distribution, with the linear constraints coming from no-signaling, measurement orthogonality, and the distribution itself. Higher levels give stricter tests but greater computational cost, so we start from low levels such as $1 + AB$.

If the SDP is infeasible at some level, the distribution cannot be produced by any quantum system. If it is feasible, the distribution lies in that level's outer approximation, which is a necessary but not sufficient condition for being quantum.

In practice, we use PyNPA, which generates the operator products for a given level, builds the moment matrix, translates $p(a, b, c \mid x, y, z)$ into linear constraints, and solves the resulting SDP with CVXPY and SDPA.

3.9 Optimization over Correlation Sets

Two types of problems are solved. The first maximizes the Bell value of p_{AB} over all tripartite distributions in a given set, with no extension, an ordinary extension, or a perfect extension.

The second fixes p_{AB} and minimizes $\bar{e}$ over all of its ordinary extensions.

For the no-signaling (NS) set, both problems are linear programs. For the quantum set, we use outer approximations given by moment matrices. Each party has projectors P_x onto outcome 0, and the moment matrix $G_{ij} = \langle w_i^\dagger w_j \rangle$ is indexed by a set of operator words $\{w_i\}$. NPA level k [**?**] uses all words of length at most k, and the almost-quantum (AQ) set [**?**] uses all words with at most one operator per party. We impose $G \succeq 0$, nonnegativity of every tripartite probability, and the extension constraints, all of which are linear in the moments. For example,

$$e_y = \langle B_y \rangle + \langle C_y \rangle - 2 \langle B_y C_y \rangle . \tag{20}$$

Because every quantum distribution lies in each of these sets, a bound obtained over a relaxation also holds for quantum theory.

3.10 Targeted Search

Random sampling explores typical states and measurements but rarely finds the small regions where nonlocality appears. We therefore also maximize the Bell value directly. For fixed real measurements, the best state in the B–C symmetric subspace is the leading eigenvector of the Bell operator restricted to that subspace. The measurement angles are then optimized from many random starting points, with Bob and Charlie sharing the same measurements so that $p_{AB} = p_{AC}$ holds exactly.

3.11 Exact Verification

Numerical solvers give approximate answers, so the key claims are verified in exact rational arithmetic. The quantum example is converted to an integer state vector and integer measurement directions, and its p_{AB} is computed exactly. For the lower bound on $\bar{e}$, the numerical dual solution of the AQ problem is rounded to rational numbers, with the positive-semidefinite multiplier written as $Z = LL^T$ so that it is exactly positive semidefinite. For any such Z, any $u \geq 0$, and any multipliers μ, every feasible moment vector v satisfies

$$\bar{e}(v) \;\geq\; \mu^T c - \sum_k |r_k|, \tag{21}$$

where r is the exactly computed residual of the dual constraints and c collects the fixed values of the constraints. This uses only the fact that every AQ moment lies in $[-1, 1]$, so the bound does not depend on solver accuracy. Finally, the perfect NS extension is found as a vertex of a linear program, and its probabilities are recomputed exactly by solving the linear system on its support.

4 Results

4.1 Random Sampling

We first applied the random-sampling procedure of Section 3 to 1,000 symmetrized Haar-random three-qubit states with Haar-random measurements ($m = 3$), with Bob and Charlie sharing the same measurements. Every sample satisfied normalization, no-signaling, and $p_{AB} = p_{AC}$ to machine precision. However, none was nonlocal: the largest value of I_{3322} was -0.304, and none was a perfect extension (the smallest mean mismatch was 0.041). When Charlie's measurements were instead sampled independently of Bob's, the marginals differed by up to 0.215, which confirms that shared measurements are necessary for $p_{AB} = p_{AC}$.

4.2 Maximum Violation Under Extension Constraints

Table 1 lists the maximum Bell violation over NS distributions. Requiring an ordinary extension lowers the maximum for every scenario, and for CGLMP it removes all violation. Requiring a perfect extension lowers it no further. These values agree with those reported in Ref. [6].

Table 1: Maximum Bell violation over no-signaling distributions.

	I_{mm22}			CGLMP (2 settings, d outcomes)			
m	NS	Ext	Perf-Ext	d	NS	Ext	Perf-Ext
2	0.5000	0.0000	0.0000	2	0.5000	0.0000	0.0000
3	1.0000	0.3333	0.3333	3	0.6667	0.0000	0.0000
4	1.5000	0.6667	0.6667	4	0.7500	0.0000	0.0000
5	2.0000	1.0000	1.0000	5	0.8000	0.0000	0.0000

Table 2 lists the corresponding bounds over moment-matrix relaxations of the quantum set. The column QB is the bound without any extension. Our almost-quantum values coincide with the Moroder level-1 values of Ref. [6], and our NPA level-3 value for $m = 3$ agrees with theirs. Our NPA level-2 values are lower than theirs. Removing the constraint that all tripartite probabilities be nonnegative reproduces their values (last row), so the difference comes from this constraint. Since every quantum distribution satisfies it, the tighter values are valid bounds.

For $m = 3$ and $m = 4$, the AQ set allows no violation with a perfect extension. For $m = 5$, however, it allows $I_{5522} = 0.0067 > 0$, again in agreement with Ref. [6]. The AQ relaxation alone therefore does not exclude perfect copying in every scenario, which is why our main result below fixes a specific distribution and certifies it directly.

4.3 An Explicit Quantum Example

The targeted search found an explicit three-qubit realization for $m = 3$, with $I_{3322} = 0.0161083$. It was reached from 14 of 40 starting points. Its checks are listed in Table 3. For $m = 4$, the

Table 2: Maximum I_{mm22} violation over moment-matrix relaxations.

Relaxation	m	QB	Ext	Perf-Ext
Almost quantum	3	0.2515	0.0398	0.0000
Almost quantum	4	0.2940	0.0650	0.0000
Almost quantum	5	–	–	0.0067*
NPA level 2	3	0.2509	0.0946	0.0900
NPA level 2	4	0.2883	0.1341	0.1239*
NPA level 2	5	–	–	–
NPA level 3	3	–	0.0273	–
NPA level 2†	3	0.2509	0.1250	0.1250

†Without nonnegativity of tripartite probabilities.

*Solver did not fully converge (constraint violation below 10^{-5}).

–: not computed in this run.

search found $I_{4422} = 0.00636$, while for $m = 5$ no positive violation was found from 60 starting points.

Table 3: Checks on the explicit three-qubit realization ($m = 3$).

Quantity	Value
I_{3322}	0.0161083
Local visibility of p_{AB}	0.98415
$\max\|p_{AB} - p_{AC}\|$	2.2×10^{-16}
Maximum no-signaling error	3.3×10^{-16}
Mismatch (e_0, e_1, e_2) of this extension	(0.2182, 0.2182, 0.3534)

In the rational version of this example, the state is an integer vector symmetric under $B \leftrightarrow C$ and each measurement direction is an integer vector. Exact arithmetic confirms $p_{AB} = p_{AC}$ and gives $I_{3322} = 0.0161082764\ldots > 0.016$, so p_{AB} is nonlocal and has an ordinary quantum extension.

4.4 Perfect Extensions of the Fixed Distribution

A comparison of maximum violations cannot tell whether a *specific* p_{AB} has a perfect extension. We therefore fixed p_{AB} from the rational example and minimized $\bar{e}$ over its extensions (Table 4).

Table 4: Minimum mean mismatch $\bar{e}$ over extensions of the fixed nonlocal p_{AB}.

Allowed extensions	Minimum $\bar{e}$
No-signaling	0 (exact perfect extension)
NPA level 2	0.0191
NPA level 3	≈ 0.223*
Almost quantum (numerical)	0.096325
Almost quantum (exact certificate)	> 0.096324
Explicit quantum extension (Table 3)	0.2632

*Solver did not fully converge (constraint violation $\sim 10^{-5}$).

The certificate of Eq. (21) gives the rigorous bound $\bar{e} > 0.096324$, with a residual correction of only 9.7×10^{-7}. Since the quantum set is contained in the AQ set, every quantum extension of this p_{AB} has $\bar{e} > 0.0963$. In contrast, the exact NS extension satisfies all 315 linear equalities and the nonnegativity of all 216 probabilities with zero error, and has $\bar{e} = 0$.

NPA level 3 contains every word of the AQ set, so it is a tighter relaxation, and it raises the minimum to about 0.223. Although this value is not certified, it indicates that the smallest mismatch achievable by any quantum extension lies roughly between 0.223 and 0.2632, the value of the explicit extension we found.

4.5 Robustness to Noise

We mixed the state with white noise, $\rho_v = v\rho + (1 - v)\mathbb{I}/8$, which gives $I_{3322}(v) = 0.01611\, v - (1 - v)$. The distribution is nonlocal for $v > 0.9841$. Table 5 shows that the minimum AQ mismatch decreases with noise but remains positive at every nonlocal point of the scan, including $v = 0.985$, where I_{3322} is below 10^{-3}. It vanishes only at $v = 0.984$, where the distribution is already local. Within numerical accuracy, the threshold for excluding a perfect AQ extension therefore coincides with the nonlocality threshold, and near it the minimum mismatch is roughly proportional to the Bell violation.

Table 5: White noise: Bell value and minimum AQ mismatch for the fixed p_{AB}.

v	I_{3322}	Minimum $\bar{e}$ (AQ)
1.000	+0.01611	0.09633
0.998	+0.01408	0.07818
0.996	+0.01204	0.06302
0.994	+0.01001	0.04990
0.992	+0.00798	0.03827
0.990	+0.00595	0.02777
0.988	+0.00391	0.01797
0.986	+0.00188	0.00855
0.985	+0.00087	0.00392
0.984	−0.00015	0.00000

4.6 Bell Violation Versus Allowed Mismatch

Finally, we allowed each e_y to be at most ε and maximized I_{3322} over AQ extensions (Table 6). The maximum is zero at $\varepsilon = 0$ and grows continuously with ε, reaching the unrestricted ordinary-extension value 0.0398 for large ε.

Table 6: Maximum I_{3322} over AQ extensions with $e_y \leq \varepsilon$.

ε	Maximum I_{3322}
0	0.00000
0.001	0.00022
0.01	0.00221
0.05	0.01088
0.1	0.02116
0.2	0.03775
0.3	0.03980

5 Discussion

5.1 Ordinary Versus Perfect Extendibility

The results show that ordinary and perfect extendibility are genuinely different. Requiring $p_{AB} = p_{AC}$ constrains the statistics Bob and Charlie each share with Alice, but it does not force them to agree on individual trials. For our example, the gap is not limited to the extension we found: the exact certificate rules out every perfect quantum extension of that p_{AB}. This answers the original question in the negative. A nonlocal distribution that admits an ordinary quantum extension need not admit a perfect quantum extension.

5.2 Dependence on the Allowed Extensions

The same p_{AB} has a perfect NS extension but no perfect quantum extension. Whether perfect copying is possible therefore depends on the theory the third party may use, not on nonlocality alone. This connects to the cryptographic motivation in Section 1: an eavesdropper limited to quantum mechanics cannot perfectly copy Bob's outcomes for this distribution, whereas a no-signaling eavesdropper can. However, $\bar{e}$ is not a key rate or an optimal guessing probability, so this is not a security proof for a complete protocol.

5.3 Methodological Lessons

Three lessons apply beyond this example. First, a random search that finds no violation does not show that none exists: all 1,000 random samples were local, yet a targeted search found a nonlocal example. Second, feasibility at a finite NPA level does not produce a quantum realization, and low levels can be loose: NPA level 2 allows $I_{3322} = 0.0900$ with a perfect extension (and 0.125 without the nonnegativity constraint), whereas the AQ set and NPA level 3 allow essentially none. Third, implementation details such as explicit probability constraints can change a relaxation's value, so they should be reported with the results.

5.4 Limitations

The value $I_{3322} \approx 0.0161$ is achievable but not proven optimal; the NPA level-3 bound 0.0273 leaves a gap. The exact certificates cover only the specific rational example, while the noise scan, the trade-off curve, and the relaxation tables are numerical. Results from the first-order solver for the largest relaxations are accurate only to about 10^{-5}.

5.5 Future Work

Several directions follow. One is to prove that the noise threshold for nonlocality and the threshold for excluding perfect quantum extensions coincide exactly, as the numerical scan suggests, and to certify points near them. Another is to test whether a fixed Bell violation forces a positive minimum mismatch for every p_{AB}, rather than for one example. Further goals are to close the gap between the explicit quantum value and the relaxation bounds, to derive a shorter analytic argument from the numerical certificate, and to extend the study to more settings and outcomes.

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